

HW5

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Plot as a function of x the pressure distribution normalized by the stagnation pressure for the attached nozzle profile. The profile is given in terms of a two column matrix where the first column is the x location and the second column is the radius of an axisymmetric circular nozzle through which the flow goes through. Assume air and CPG.

1. Case 1: fully subsonic, you may choose a back pressure
2. Case 2: fully subsonic, with the throat at sonic, what is the back pressure?
3. Case 3: Normal shock at $x = 0.75$ in the nozzle, what is the back pressure?
4. Case 4: Normal shock at the exit of the nozzle, what is the back pressure?
5. Case 5: Ideally expanded, what is the back pressure?
6. What pressure range corresponds to oblique shockwaves at the exit of the nozzle?
7. What pressure range corresponds to expansion waves at the exit of the nozzle?

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In [63]: import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import brentq

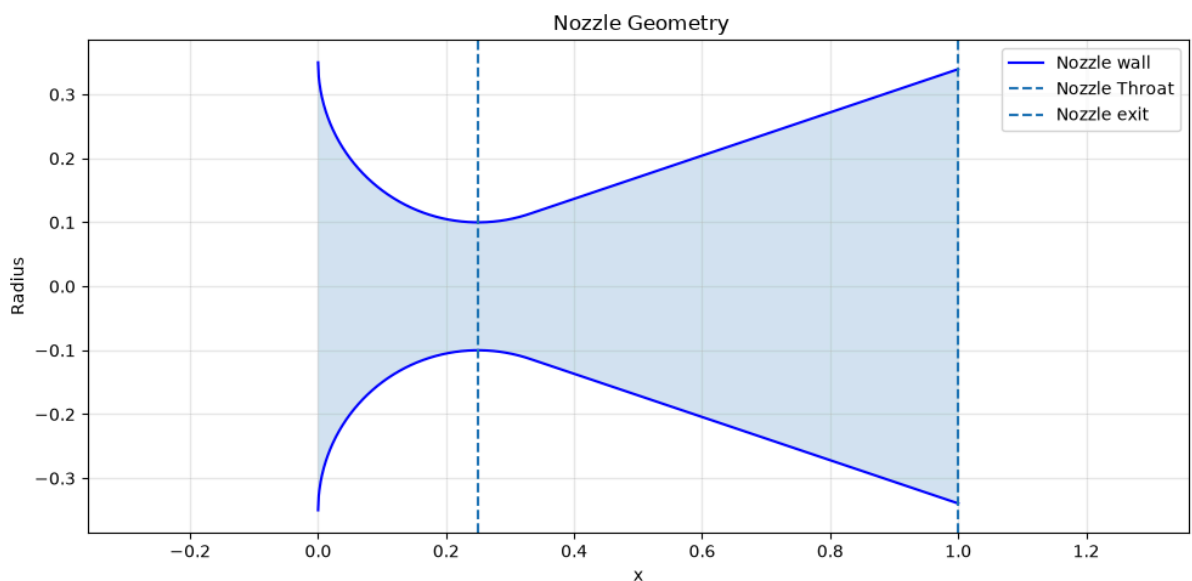
gamma = 1.4
Nozzle = np.loadtxt("Pnozzle.txt", delimiter=",")
Nozzle_Geom = []

x = Nozzle[:, 0]
r = Nozzle[:, 1]
A = r * r * np.pi

Nozzle_Geom.append([x, r, A])
A_throat= A[np.argmin(A)]
A_e = A[-1]

plt.figure(figsize=(10, 5))
plt.plot(x, r, color = 'b', label='Nozzle wall')
plt.plot(x, -r, color = 'b' )
plt.fill_between(x, -r, r, alpha = 0.2)
plt.axvline( x=x[A == A_throat], linestyle='--', label='Nozzle Throat')
plt.axvline( x=x[A == A_e], linestyle='--', label='Nozzle exit')
plt.xlabel("x")
plt.ylabel("Radius")
plt.title(" Nozzle Geometry")
plt.axis("equal")
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()

```



$$\left(\frac{A}{A^*}\right)^2 = \frac{1}{M^2} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M^2 \right) \right]^{(\gamma+1)/(\gamma-1)} \quad (5.20)$$

Equation (5.20) is called the *area-Mach number relation*, and it contains a striking result. Turned inside out, Eq. (5.20) tells us that $M = f(A/A^*)$, i.e., the Mach number at any location in the duct is a function of the *ratio* of the local duct area to the sonic throat area. As seen from Eq. (5.15), A must be greater than or at least equal to A^* ; the case where $A < A^*$ is physically not possible in an isentropic flow. Also, from Eq. (5.20) there are two values of M that correspond to a given $A/A^* > 1$, a subsonic and a supersonic value. The solution of Eq. (5.20) is plotted in Fig. 5.13,

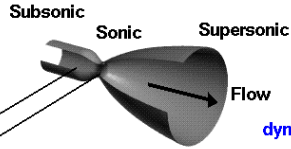


Isentropic Flow

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Mach = M
speed of sound = a
gas constant = R
specific heat ratio = γ

t = total conditions
* = sonic conditions



velocity = v
pressure = p
temperature = T
density = ρ
area = A
dynamic pressure = q

$$\begin{aligned} (1) \quad M &= \frac{v}{a} & (6) \quad \frac{p}{p_t} &= \left(1 + \frac{\gamma-1}{2} M^2 \right)^{-\frac{\gamma}{\gamma-1}} \\ (2) \quad a &= \sqrt{\gamma \frac{p}{\rho}} = \sqrt{\gamma RT} & (7) \quad \frac{T}{T_t} &= \left(1 + \frac{\gamma-1}{2} M^2 \right)^{-1} \\ (3) \quad \frac{p}{\rho^\gamma} &= \text{Constant} = \frac{p_t}{\rho_t^\gamma} & (8) \quad \frac{\rho}{\rho_t} &= \left(1 + \frac{\gamma-1}{2} M^2 \right)^{-\frac{1}{\gamma-1}} \\ (4) \quad \frac{p}{p_t} &= \left(\frac{\rho}{\rho_t} \right)^\gamma = \left(\frac{T}{T_t} \right)^{\frac{\gamma}{\gamma-1}} & (9) \quad \frac{A}{A^*} &= \left(\frac{\gamma+1}{2} \right)^{-\frac{\gamma+1}{2(\gamma-1)}} \frac{\left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma+1}{2(\gamma-1)}}}{M} \\ (5) \quad q &= \frac{1}{2} \rho v^2 = \frac{\gamma}{2} p M^2 \end{aligned}$$

```

In [ ]: #MachFromArea function takes in the Area ratio, gamma, type
def MachFromArea (A_ratio, gamma, type):

    #This is the 'sub-function' that returns Area ratio subtracted by the
    given Area ratio to be optimized by Brentq, where solution will be zero.
    def f (m_i):
        Relation = (1/(m_i ** 2)) * ((2/(gamma +1)) * (1 + ((gamma - 1)/2 *
(m_i ** 2)))) * ((gamma+1)/(gamma-1))
        return Relation - A_ratio ** 2

    if type == "subsonic":
        return brentq(f, 0.001, 1)
    else:
        return brentq(f, 1, 10)

# I verified Using the NASA Isentropic Flow calculator :
https://www.grc.nasa.gov/www/k-12/airplane/isentrop.html
print(MachFromArea(1.1, 1.4, "subsonic"))
print(MachFromArea(2.0, 1.4, "supersonic"))

#The function finds Mach number from the given pressure ratio. It sweeps
only from 0.528 to 1, to not account for the supersonic case.
def MachFromPressure(P_ratio, gamma):
    def f(m_i):
        M = m_i ** 2
        Relation = (1 + ((gamma-1)/2)*M) **((-gamma)/(gamma-1))
        return Relation - P_ratio
    return brentq(f,0, 100)

def PressureRatio(A_ratio, gamma, type):
    m = MachFromArea(A_ratio, gamma, type)
    M = m * m
    P_ratio = (1 + ((gamma - 1)/2) * M) ** (-gamma / (gamma - 1))
    return P_ratio

def AreaRatio(m_i, gamma):
    M = m_i ** 2
    A_ratio = np.sqrt(1/M * ((2/(gamma+1))*(1 + ((gamma-1)/2) * M)) *
((gamma+1)/(gamma-1)))
    return A_ratio

0.6923991394638542
2.1971981216521863

```

Part A

Fully subsonic case, with arbitrary back pressure. I chose 99.9% of critical back pressure

1. Calculate the imaginary area Astar where M=1 as a ratio to A_e

2. Calculate the Mach number of $A_{\text{given}} / \text{Imaginary } A_{\text{star}}$
3. Calculate the normalized pressure ratio based on the area ratio found in step 2.
4. Plot the results

```

In [199]: gamma = 1.4
          Pb_ratio = 0.999

          # Subsonic Mach number at the nozzle exit
          M_exit = MachFromPressure(Pb_ratio, gamma)

          # Exit area / critical area
          A_ratio_e = AreaRatio(M_exit, gamma)

          # Estimate critical area
          Astar = A[-1] / A_ratio_e
          print("Area Ratio of Imaginary Nozzle Astar", A_ratio_e)
          print("Exit Mach number:", M_exit)
          print("Critical area A*:", Astar)

          p_x = np.array([PressureRatio(Ai / Astar, gamma, "subsonic") for Ai in A])

          print(f"Back pressure ratio:", Pb_ratio)

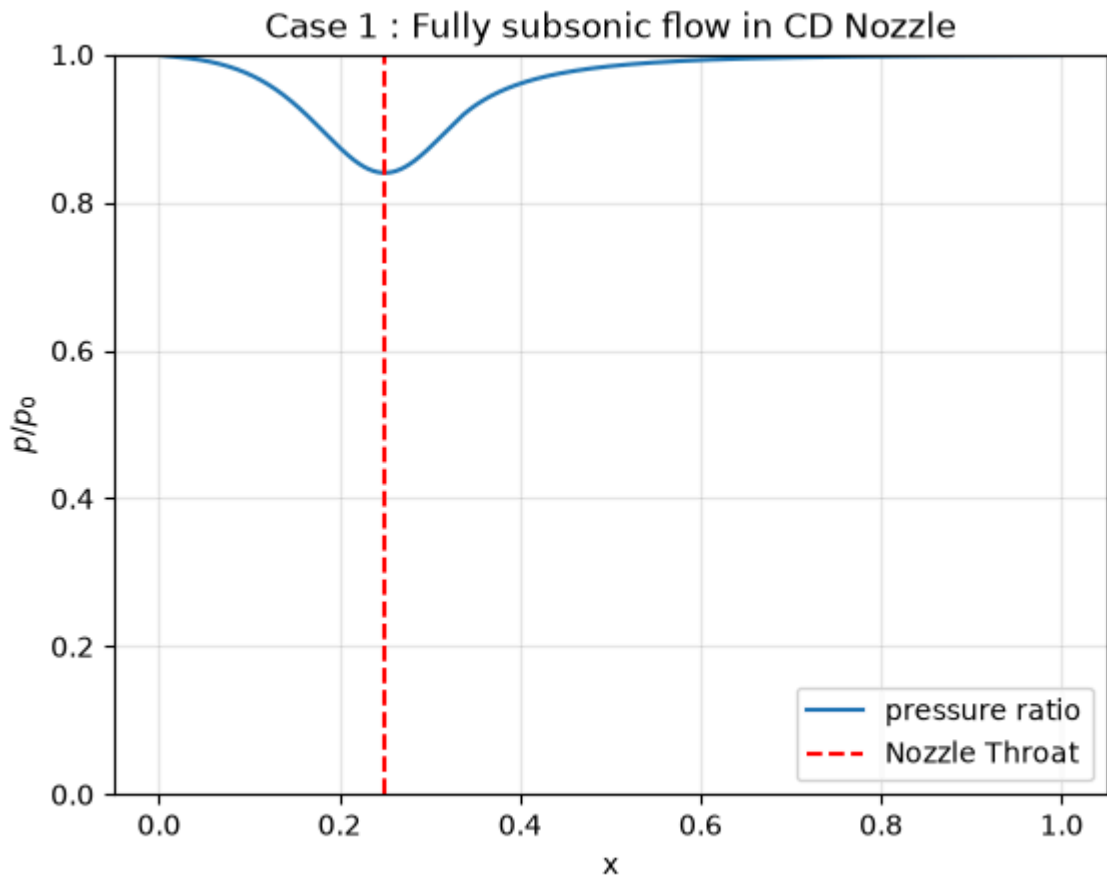
          plt.plot(x, p_x, label=f"pressure ratio")
          plt.axvline( x=x[A == A_throat], linestyle='--',color = 'red', label='Nozzle Throat')
          plt.xlabel("x")
          plt.ylabel("$p/p_0$")
          plt.grid(True, alpha=0.3)
          plt.ylim(0, 1)
          plt.legend(loc = 'lower right')
          plt.figtext(0.2, -0.02,"Back Pressure Ratio = 0.999",ha="center",fontsize=10)
          plt.title("Case 1 : Fully subsonic flow in CD Nozzle")
          plt.show()

```

```

Area Ratio of Imaginary Nozzle Astar 15.319269778894412
Exit Mach number: 0.03780860346801294
Critical area A*: 0.023629287097654297
Back pressure ratio: 0.999

```



PART B

Fully subsonic, with the throat at sonic, what is the back pressure?

1. For the full subsonic flow with only the throat at sonic, $A_{\text{throat}} = A_{\text{star}}$.
2. Calculate the mach number of $A_{\text{given}} / A_{\text{star}}$ area ratio.
3. Calculate the normalized pressure ratio based on the area ratio found in step 2.
4. Plot the results.

```

In [200]: Nozzle_ratio = A_e/A_throat

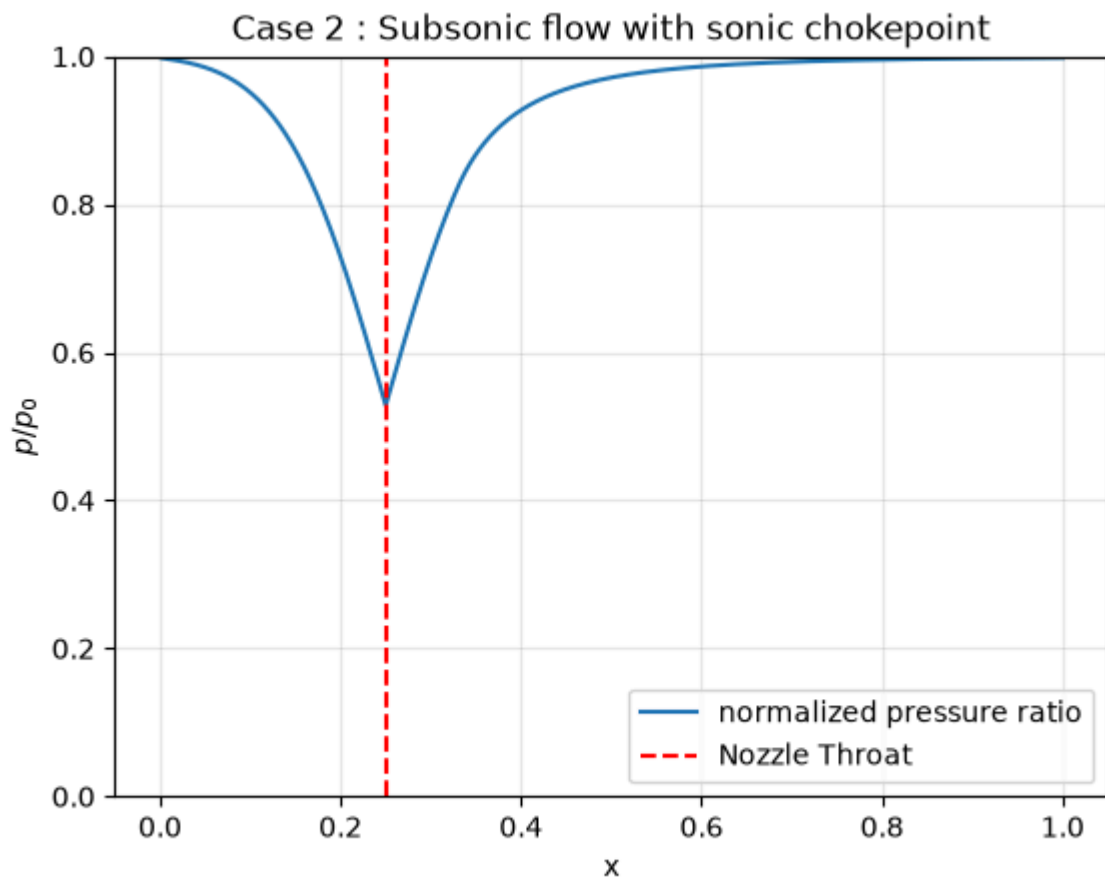
p_x = np.array([PressureRatio(Ai / A_throat, gamma, "subsonic") for Ai in
A])

print(f"Back pressure ratio:", )

plt.plot(x, p_x, label=f"normalized pressure ratio")
plt.axvline( x=x[A = A_throat], linestyle='--',color = 'red', label='Nozzle
Throat')
plt.xlabel("x")
plt.ylabel("$p/p_0$")
plt.grid(True, alpha=0.3)
plt.ylim(0, 1)
plt.title("Case 2 : Subsonic flow with sonic chokepoint")
plt.figtext(0.2, -0.02,f"Back Pressure Ratio =
{p_x[-1]:.4f}",ha="center",fontsize=10)
plt.legend()
plt.show()

```

Back pressure ratio:



Back Pressure Ratio = 0.9982

PART C

Case 3: Normal shock at $x = 0.75$ in the nozzle, what is the back pressure?

Normal Shock Wave



Zone 0 Upstream	γ – specific heat ratio	Shock Wave	Zone 1 Downstream
p – static pressure			p – static pressure
p_t – total pressure			p_t – total pressure
T – static temperature			T – static temperature
T_t – total temperature			T_t – total temperature
ρ – density			ρ – density
M – Mach number			M_1 – Mach number



$$\frac{p_1}{p_0} = \frac{2\gamma M^2 - (\gamma - 1)}{\gamma + 1}$$

$$\frac{p_{t1}}{p_{t0}} = \left[\frac{(\gamma + 1) M^2}{(\gamma - 1) M^2 + 2} \right]^{\frac{\gamma}{\gamma - 1}} \left[\frac{(\gamma + 1)}{2\gamma M^2 - (\gamma - 1)} \right]^{\frac{1}{\gamma - 1}}$$

$$\frac{T_1}{T_0} = \frac{[2\gamma M^2 - (\gamma - 1)] [(\gamma - 1) M^2 + 2]}{(\gamma + 1)^2 M^2}$$

$$\frac{T_{t1}}{T_{t0}} = 1$$

$$\frac{\rho_1}{\rho_0} = \frac{(\gamma + 1) M^2}{(\gamma - 1) M^2 + 2}$$

$$M_1^2 = \frac{(\gamma - 1) M^2 + 2}{2\gamma M^2 - (\gamma - 1)}$$

www.nasa.gov

- Up until $x = 0.75$, the normalized pressure looks exactly like Case 5, ideally expanded.
- At $x = 0.75$, using the $A(x=0.75)$, find the Mach number.
- Construct the imaginary A_{star2} like case 1
- Using the found mach number, go through the normal shock to find the M_2 after the NS and the pressure ratio after the Normal Shock

$$5. \quad \frac{p_e}{p_{01}} = \left(\frac{p_e}{p_{02}} \right) \left(\frac{p_{02}}{p_{01}} \right)$$

```
In [191]: # This function calculates the mach after the normal shock using the Normal
           Shock relations
           def NSMach(m_1, gamma):
               M = m_1 ** 2
               Mach = ((gamma - 1) * M + 2) / (2 * gamma * M - (gamma - 1))
               return np.sqrt(Mach)

           #This function calculates the pressure ratio P2/P1 after the normal shock
           using the Normal Shock relations
           def NSPressure(m_1, gamma):
               M = m_1 ** 2
               return (2 * gamma * M - (gamma - 1)) / (gamma + 1)

           #This function calculates the stag pressure ratio
           def NSStagPressure(m_1, gamma):
               M = m_1 ** 2
               relation = (((gamma + 1) * M) / ((gamma - 1) * M + 2)) ** (gamma / (gamma
               - 1)) * ((gamma + 1) / ((2 * gamma * M) - (gamma - 1))) ** (1 / (gamma - 1))
               return relation

           print(NSMach(2, 1.4))
           print(NSPressure(2, 1.4))
           print(NSStagPressure(2, 1.4))
```

0.5773502691896257

4.5

0.7208738614847455

```

In [201]: i_throat = np.argmin(A)
i_shock = np.searchsorted(x, 0.75)

p_x = np.zeros(len(A))
gamma = 1.4

m_1 = MachFromArea((A[i_shock])/A[i_throat]), gamma, 'supersonic')
m_2 = NSMach(m_1, gamma)
A_star2 = A[i_shock] / AreaRatio(m_2, gamma)
StagPressure = NSStagPressure(m_1, gamma)

for i in range (0, len(A)):
    if i < i_throat :
        p_x[i] = PressureRatio((A[i]/A[i_throat]), gamma, 'subsonic')
    elif i<i_shock:
        p_x[i] = PressureRatio((A[i]/A[i_throat]), gamma, 'supersonic')
    elif i ≥ i_shock:
        p_x[i] = (StagPressure * PressureRatio((A[i]/A_star2), gamma,
'subsonic'))

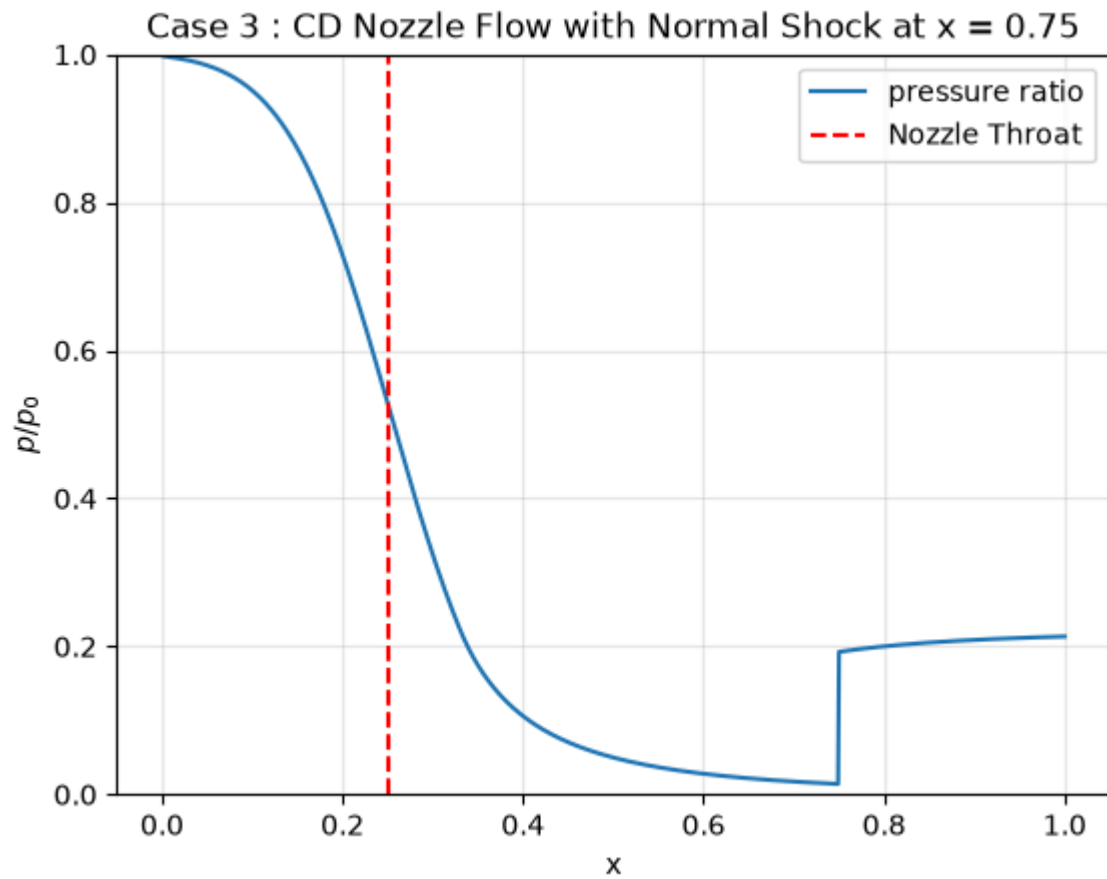
        #Need to find the imaginary throat after the NS, A_star2

#Printing Mach number right before the NS
#Printing Area right before the NS
print(A[i_shock])

plt.plot(x, p_x, label=f"pressure ratio")
plt.xlabel("x")
plt.ylabel("$p/p_0$")
plt.axvline( x=x[A == A_throat], linestyle='--',color = 'red', label='Nozzle
Throat')
plt.grid(True, alpha=0.3)
plt.ylim(0, 1)
plt.title("Case 3 : CD Nozzle Flow with Normal Shock at x = 0.75")
plt.legend()
plt.figtext(0.2, -0.02,f"Back Pressure Ratio =
{p_x[-1]:.4f}",ha="center",fontsize=10)
plt.show()

```

0.2042899541409645



PART D

Case 4 : Normal Shock at the nozzle exit. What is the back pressure?

```

In [202]: i_throat = np.argmin(A)
i_shock = np.searchsorted(x, 1)

p_x = np.zeros(len(A))
gamma = 1.4

m_1 = MachFromArea((A[i_shock])/A[i_throat]), gamma, 'supersonic')
m_2 = NSMach(m_1, gamma)
A_star2 = A[i_shock] / AreaRatio(m_2, gamma)
StagPressure = NSStagPressure(m_1, gamma)

for i in range (0, len(A)):
    if i < i_throat :
        p_x[i] = PressureRatio((A[i]/A[i_throat]), gamma, 'subsonic')
    elif i<i_shock:
        p_x[i] = PressureRatio((A[i]/A[i_throat]), gamma, 'supersonic')
    elif i ≥ i_shock:
        p_x[i] = (StagPressure * PressureRatio((A[i]/A_star2), gamma,
'subsonic'))

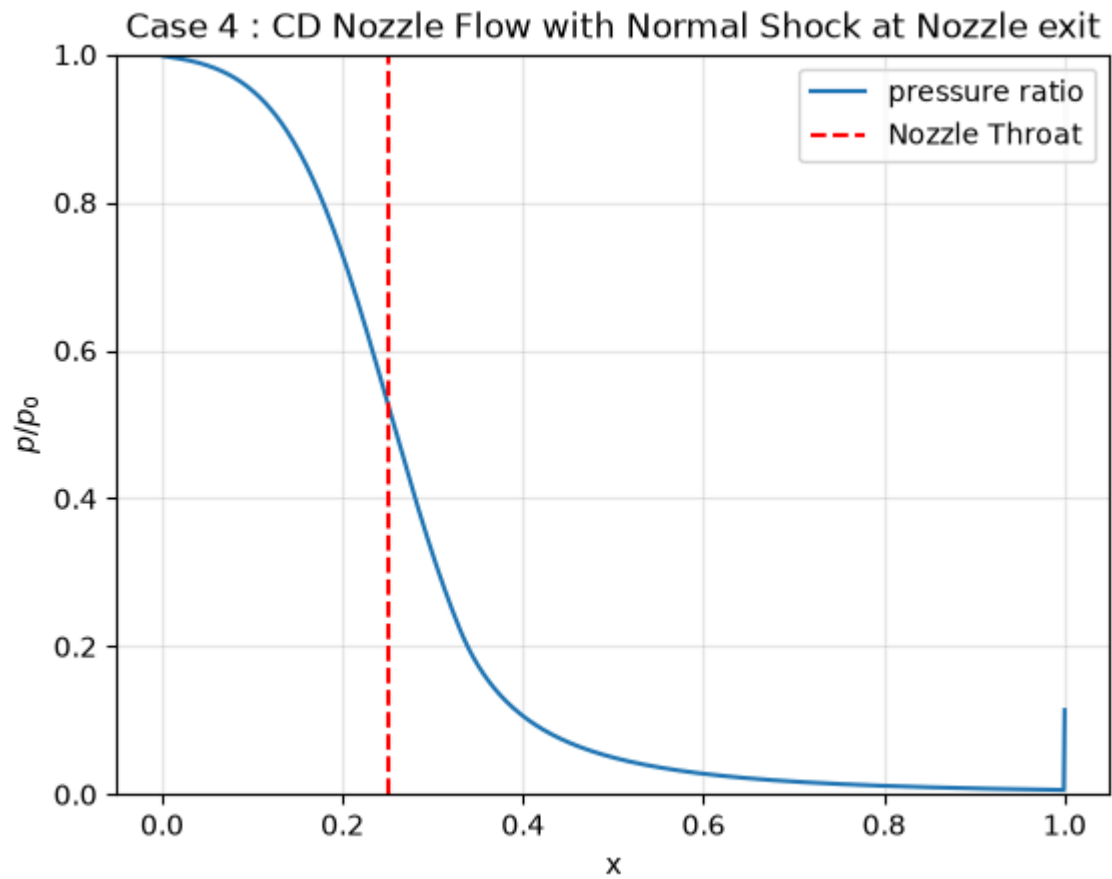
        #Need to find the imaginary throat after the NS, A_star2

#Printing Mach number right before the NS
#Printing Area right before the NS
print(A[i_shock])

plt.plot(x, p_x, label=f"pressure ratio")
plt.xlabel("x")
plt.ylabel("$p/p_0$")
plt.axvline( x=x[A == A_throat], linestyle='--',color = 'red', label='Nozzle
Throat')
plt.grid(True, alpha=0.3)
plt.ylim(0, 1)
plt.legend()
plt.title("Case 4 : CD Nozzle Flow with Normal Shock at Nozzle exit")
plt.figtext(0.2, -0.02,f"Back Pressure Ratio =
{p_x[-1]:.4f}",ha="center",fontsize=10)
plt.show()

```

0.3619834237319151



PART E

Case 5 : Ideally Expanded. What is the back pressure?

```

In [203]: Nozzle_ratio = A_e/A_throat
x_throat = np.argmin(A)

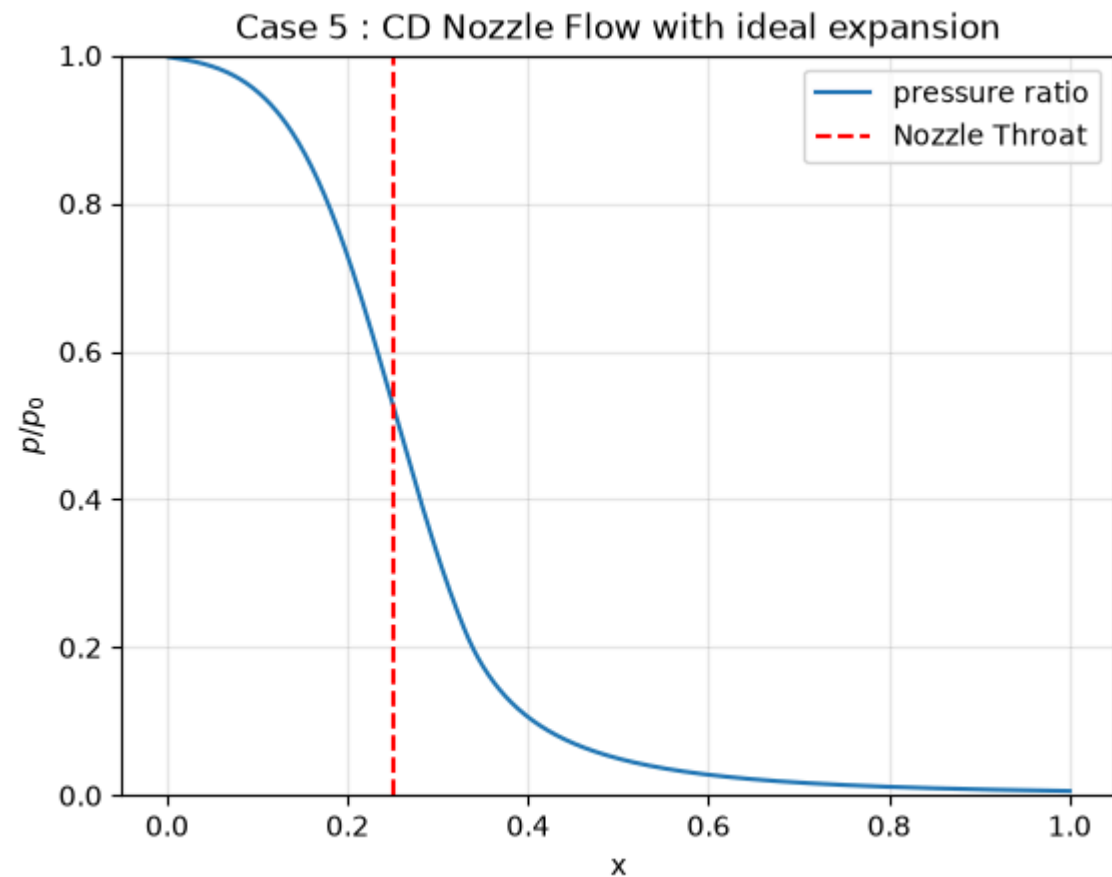
p_x = np.zeros(len(A))
for i in range (0, len(A)):
    if i < x_throat :
        p_x[i] = PressureRatio((A[i]/A[x_throat]), gamma, 'subsonic')
    else :
        p_x[i] = PressureRatio((A[i]/A[x_throat]), gamma, 'supersonic')

print(p_x[-1])

plt.plot(x, p_x, label=f"pressure ratio")
plt.xlabel("x")
plt.ylabel("$p/p_0$")
plt.axvline( x=x[A == A_throat], linestyle='--',color = 'red', label='Nozzle Throat')
plt.grid(True, alpha=0.3)
plt.ylim(0, 1)
plt.legend()
plt.title("Case 5 : CD Nozzle Flow with ideal expansion")
plt.figtext(0.2, -0.02,f"Back Pressure Ratio = {p_x[-1]:.4f}",ha="center",fontsize=10)
plt.show()

```

0.005912985826544639



Part F

What pressure range corresponds to oblique shockwaves at the exit of the nozzle?

The Pressure difference between the Exit pressure in the Part D, Normal shock at the end of the nozzle, to the exit pressure of the ideally expanded nozzle corresponds to oblique shockwaves at the exit of the nozzle. Thus,

$$0.0059 \leq \frac{P_b}{P_0} \leq 0.1139$$

corresponds to the expansion wave.

Part G

What pressure range corresponds to expansion waves at the exit of the nozzle

The pressure range from 0 to the back pressure ratio of the ideally expanded nozzle corresponds to the expansion wave regime. Thus, the back pressure ratio range

$$0 \leq \frac{P_b}{P_0} \leq 0.0059$$

corresponds to the expansion wave.